

6.1.25. Let $R_i = \{x \in \mathbb{R} \mid 1 \leq x \leq 1 + (1/i)\} = [1, 1 + (1/i)]$ for each positive integer i .

a. $\bigcup_{i=1}^4 R_i = ?$

b. $\bigcap_{i=1}^4 R_i = ?$

c. Are $R_1, R_2, R_3, \dots$ mutually disjoint? Explain.

d. $\bigcup_{i=1}^n R_i = ?$

e. $\bigcap_{i=1}^n R_i = ?$

f. $\bigcup_{i=1}^{\infty} R_i = ?$

g. $\bigcap_{i=1}^{\infty} R_i = ?$

a. $\bigcup_{i=1}^4 R_i = ?$

Let $R_i = \{x \in \mathbb{R} \mid 1 \leq x \leq 1 + (1/i)\} = [1, 1 + (1/i)]$

Since $i \in \mathbb{Z}^+$

$$\bigcup_{i=1}^4 R_i = R_1 \cup R_2 \cup R_3 \cup R_4$$

$$R_1 = \{x \in \mathbb{R} \mid 1 \leq x \leq 1 + (1/1)\}$$

$$R_2 = \{x \in \mathbb{R} \mid 1 \leq x \leq 1 + (1/2)\}$$

$$R_3 = \{x \in \mathbb{R} \mid 1 \leq x \leq 1 + (1/3)\}$$

$$R_4 = \{x \in \mathbb{R} \mid 1 \leq x \leq 1 + (1/4)\}$$

Definition of union of an indexed collection:

$$\bigcup_{i=1}^4 A_i = \{x \in U \mid x \in A_i \text{ for at least one } i = 1, 2, 3, 4\}$$

Therefore, $\bigcup_{i=1}^4 R_i = \{x \in \mathbb{R} \mid 1 \leq x \leq 1 + (1/1)\}$

Solution: $\bigcup_{i=1}^4 R_i = \{x \in \mathbb{R} \mid 1 \leq x \leq 1 + (1/1)\}$

b. $\bigcap_{i=1}^4 R_i = ?$

Let $R_i = \{x \in \mathbb{R} \mid 1 \leq x \leq 1 + (1/i)\} = [1, 1 + (1/i)]$

Since $i \in \mathbb{Z}^+$

$$\bigcap_{i=1}^4 R_i = R_1 \cap R_2 \cap R_3 \cap R_4$$

$$R_1 = \{x \in \mathbb{R} \mid 1 \leq x \leq 1 + (1/1)\}$$

$$R_2 = \{x \in \mathbb{R} \mid 1 \leq x \leq 1 + (1/2)\}$$

$$R_3 = \{x \in \mathbb{R} \mid 1 \leq x \leq 1 + (1/3)\}$$

$$R_4 = \{x \in \mathbb{R} \mid 1 \leq x \leq 1 + (1/4)\}$$

Definition of intersection of an indexed collection:

$$\bigcap_{i=1}^4 A_i = \{x \in U \mid x \in A_i \text{ for every } i = 1, 2, 3, 4\}$$

Therefore, $\bigcap_{i=1}^4 R_i = \{x \in \mathbb{R} \mid 1 \leq x \leq 1 + (1/4)\}$

Solution: $\bigcap_{i=1}^4 R_i = \{x \in \mathbb{R} \mid 1 \leq x \leq 1 + (1/4)\}$

c. Are $R_1, R_2, R_3, \dots$ mutually disjoint? Explain.

Definition of mutually disjoint:

$$A_i \cap A_j = \emptyset \text{ whenever } i \neq j$$

Since $R_1 \supseteq R_2 \supseteq R_3, \dots, \supseteq R_n$

So by definition, we must show whether the sets have no elements in common.

However, $R_1 \supseteq R_2 \supseteq R_3, \dots, \supseteq R_n$, hence they are not mutually disjoint, because R_1 contains elements from all other sets.

Solution: $R_1, R_2, R_3, \dots$ are not mutually disjoint.

d. $\bigcup_{i=1}^n R_i = ?$

Let $R_i = \{x \in \mathbb{R} \mid 1 \leq x \leq 1 + (1/i)\} = [1, 1 + (1/i)]$

$$\bigcup_{i=1}^n R_i = R_1 \cup R_2 \cup R_3 \dots \cup R_n$$

Definition of union of an indexed collection:

$$\bigcup_{i=1}^n A_i = \{x \in U \mid x \in A_i \text{ for at least one } i = 1, 2, 3, \dots, n\}$$

Therefore, $\bigcup_{i=1}^n R_i = \{x \in \mathbb{R} \mid 1 \leq x \leq 1 + (1/1)\}$

Solution: $\bigcup_{i=1}^n R_i = \{x \in \mathbb{R} \mid 1 \leq x \leq 1 + (1/1)\}$

e. $\bigcap_{i=1}^n R_i = ?$

Let $R_i = \{x \in \mathbb{R} \mid 1 \leq x \leq 1 + (1/i)\} = [1, 1 + (1/i)]$

$$\bigcap_{i=1}^n R_i = R_1 \cap R_2 \cap R_3 \dots \cap R_n$$

Definition of intersection of an indexed collection:

$$\bigcap_{i=1}^n A_i = \{x \in U \mid x \in A_i \text{ for every } i = 1, 2, 3, \dots, n\}$$

Therefore, $\bigcap_{i=1}^n R_i = \{x \in \mathbb{R} \mid 1 \leq x \leq 1 + (1/n)\}$

Solution: $\bigcap_{i=1}^n R_i = \{x \in \mathbb{R} \mid 1 \leq x \leq 1 + (1/n)\}$

f. $\bigcup_{i=1}^{\infty} R_i = ?$

Let $R_i = \{x \in \mathbb{R} \mid 1 \leq x \leq 1 + (1/i)\} = [1, 1 + (1/i)]$

$$\bigcup_{i=1}^{\infty} R_i = R_1 \cup R_2 \cup R_3 \dots$$

Definition of union of an indexed collection:

$$\bigcup_{i=1}^{\infty} A_i = \{x \in U \mid x \in A_i \text{ for at least one positive integer } i\}$$

Therefore, $\bigcup_{i=1}^{\infty} R_i = \{x \in \mathbb{R} \mid 1 \leq x \leq 1 + (1/1)\}$

Solution: $\bigcup_{i=1}^{\infty} R_i = \{x \in \mathbb{R} \mid 1 \leq x \leq 1 + (1/1)\}$

g. $\bigcap_{i=1}^{\infty} R_i = ?$

Let $R_i = \{x \in \mathbb{R} \mid 1 \leq x \leq 1 + (1/i)\} = [1, 1 + (1/i)]$

$\bigcap_{i=1}^{\infty} R_i = R_1 \cap R_2 \cap R_3 \dots$

Definition of intersection of an indexed collection:

$$\bigcap_{i=1}^{\infty} A_i = \{x \in U \mid x \in A_i \text{ for every positive integer } i\}$$

Since $R_i = [1, 1 + (1/i)]$, as i gets larger:

$$R_1 = [1, 1 + (1/1)] = [1, 2]$$

$$R_2 = [1, 1 + (1/2)] = [1, 1.5]$$

$$R_3 = [1, 1 + (1/3)] = [1, 1.33\dots]$$

$$R_{10} = [1, 1 + (1/10)] = [1, 1.1]$$

$$R_{100} = [1, 1 + (1/100)] = [1, 1.01]$$

$$R_{1000} = [1, 1 + (1/1000)] = [1, 1.001]$$

Hence, the interval keeps shrinking toward 1, and there is exactly one number that is in every set.

Therefore, $\bigcap_{i=1}^{\infty} R_i = \{1\}$.

Solution: $\bigcap_{i=1}^{\infty} R_i = \{1\}$